

# Site Adaptation of Satellite-Derived GHI Estimates in Northwestern Argentina

Rubén Ledesma  Germán Salazar 

**Abstract**—This study evaluates the accuracy of satellite-based and reanalysis-based estimates of global horizontal irradiance (GHI) in northwestern Argentina through the application of site-adaptation techniques. Four GHI estimation models are analyzed—two satellite-based (CAM5 Heliosat-4 and LSA-SAF MDSSTD) and two meteorological reanalysis-based (MERRA-2 and ERA-5)—across five representative locations in the region, encompassing various altitudes and climatic conditions. Three adaptation approaches are applied: traditional linear correction, adjustment based on the error series, and adaptation using deseasonalized errors. Results show that all models improve significantly after adaptation, with error-based approaches—particularly the deseasonalized method—yielding the largest reductions in root mean square error (RMSE). Furthermore, satellite models initially outperform reanalysis models; however, the latter benefit considerably from adaptation. The study concludes that site adaptation is a key tool for improving solar resource estimation, especially in regions with limited measurement infrastructure and complex topography, such as northwestern Argentina.

**Index Terms**—Global Horizontal Solar Irradiance, Satellite Models, Reanalysis Data, Solar Resource Assessment, Northwest Argentina.

## I. INTRODUCTION

ACCURATE characterization of the solar resource at a prospective project site is fundamental for the simulation and financial evaluation of solar energy systems. Among the various sources of uncertainty, variability in solar irradiance data is the predominant factor impacting the economic risk assessment of utility-scale photovoltaic (PV) installations [1], [2]. Ideally, engineering and financial analyses should rely on a minimum of ten years of on-site solar irradiance records, verified through an established quality assurance protocol. However, such long-term, high-quality ground-based measurements are seldom available, as it is uncommon for consistent data collection to have occurred near the designated project site. In this context, “near” typically refers to within 15–20 km of the project location—beyond which satellite-derived estimates generally outperform spatial extrapolations or interpolations of ground-based data [3], [4].

To address the lack of direct measurements, solar resource assessments often utilize modeled data obtained from satellite imagery or numerical weather prediction models, commonly referred to as satellite and reanalysis datasets, respectively. These modeled datasets are widely available—some free of

charge, others commercial—and differ in their spatial and temporal resolutions. Despite their availability, these data sources are subject to significant uncertainties, which can affect the projected return on investment and thus the medium-term feasibility of solar energy projects [5], [6]. For this reason, it is essential to validate such datasets against high-quality ground measurements in order to identify systematic biases and determine the most appropriate data source for a given region. The accuracy and applicability of a modeled dataset within a climatically homogeneous zone are typically evaluated by comparing its outputs against concurrent observations collected at ground stations within that zone [7]–[11].

Satellite-based models rely on high-frequency, moderate-resolution imagery from geostationary satellites, enabling consistent monitoring of cloud cover across broad geographical areas with update intervals ranging from 10 to 15 minutes and spatial resolutions between 500 and 1000 meters per pixel. Given that large PV plants (e.g., those exceeding 10 MW of installed capacity) may span multiple satellite pixels, these models are well-suited for estimating incident irradiance on PV modules and thus predicting energy yield [6]. A distinct advantage of satellite models is their use of actual observed cloud data, in contrast to the synthetic cloud cover data generated by numerical models. This characteristic renders satellite-derived products particularly effective for solar resource assessment via remote sensing techniques [9], [12]–[14]. Consequently, satellite datasets are generally preferred over reanalysis data for resource assessment purposes, though this preference should be substantiated by site-specific ground measurements, especially in regions characterized by complex terrain.

The northwestern region of Argentina presents significant potential for the analysis of solar resources using satellite-based datasets, owing to several compelling factors. First, the region exhibits an abundance of solar irradiance, particularly in its western zones, where levels consistently exceed the global average. Additionally, the complex geographical variability and pronounced elevation gradients contribute to the formation of diverse microclimates within relatively short distances—ranging from tens to hundreds of kilometers. These microclimatic differences substantially influence the spatial distribution, variability, and average levels of solar radiation. They also result in considerable heterogeneity in surface albedo conditions, which are known to affect the accuracy of satellite-derived solar irradiance estimates [15]–[17].

Moreover, the region suffers from a limited network of operational radiometric stations equipped with traceable sensors, rigorous quality control, and consistent maintenance. In prior

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studies evaluating the performance of solar irradiance estimation models in northwestern Argentina [18]–[20], significant discrepancies have been reported between modeled data and ground-based measurements. These findings underscore the need for post-processing or adjustment procedures aimed at minimizing the error between modeled and observed values to improve the reliability of satellite-derived datasets in this region.

When ground-based measurements are available over a period that overlaps temporally with the modeled dataset at a specific location, it is possible to adapt satellite-derived solar irradiance values to better reflect local conditions [21], [22]. The objective is to derive an adaptation function that systematically improves the agreement between satellite estimates and ground observations. It is important to note that this process is referred to as "adaptation" rather than "correction," as it does not enforce exact equivalence between modeled and measured values. Instead, it seeks to reduce systematic discrepancies while preserving the statistical integrity of the dataset. This approach is commonly known as Site Adaptation (SA). Once the adaptation function is established, it can be applied retrospectively to the entire modeled dataset outside the overlap period, thereby extending the value of limited ground measurements to enhance the accuracy of long-term satellite-based solar resource assessments.

Recent studies have evaluated the performance of Site Adaptation (SA) techniques in the northwestern region of Argentina. These investigations have explored a range of methodologies, from simple linear regression models to more advanced machine learning approaches [23], [24]. In both cases, a conventional strategy has been employed, wherein the modeled global horizontal irradiance ( $GHI_{modeled}$ ) is adjusted based on ground-measured global horizontal irradiance ( $GHI_{measured}$ ). Alternative approaches to this conventional method also exist. For example, the study presented in [25] proposes adapting the time series formed by the residuals, i.e., the difference between  $GHI_{measured}$  and  $GHI_{modeled}$ , as a means to improve the accuracy of modeled estimates.

This study aims to evaluate Site Adaptation (SA) methodologies for northwestern Argentina by establishing a reflective benchmark using several freely available hourly modeled solar irradiance datasets. Five locations, each representing distinct environmental and climatic conditions within the region, are considered: Yuto (Yu, 401 m above sea level), Salta (Sa, 1,233 m), San Carlos (Sca, 1,624 m), El Rosal (Ero, 3,355 m), and La Quiaca (Lq, 3,450 m). Due to their diverse geographical characteristics, these sites exhibit significantly different climate patterns and cloud cover behavior. For instance, while clear-sky conditions are predominant in Ero and Lq, Sa and Sca experience a wider range of cloudy conditions. These latter sites pose unique challenges for remotely estimating solar irradiance, albeit for different reasons. Collectively, the selected locations capture a representative spectrum of the meteorological and topographical diversity found in the region. The primary contribution of this work is the classification and comparative evaluation of model performance across these sites under a unified framework, thereby providing insight into the typical accuracy achievable through SA techniques

in northwestern Argentina.

The remainder of this article is organized as follows. Section II describes the sites, the measured data and the quality control procedures used. Section III presents the satellite and reanalysis data sets used for site adaptation approaches. Section IV gives a brief introduction to the evaluation metrics, and Section V presents the results and discussion. Finally, Section VI summarises our conclusions.

## II. GROUND MEASUREMENTS

The sites considered in this study are summarized in Table I, including their identification codes, geographic coordinates, elevation above sea level, measurement periods, Köppen–Geiger climate classification [26], and the type of pyranometer used. The stations are distributed across northwestern Argentina and represent a range of climatic and topographic conditions, from subtropical lowlands to high-altitude Andean plateaus. The Sa station is located on the experimental campus of INENCO at the National University of Salta, in a pre-Andean urban setting within the Lerma Valley. This area frequently experiences cloud formation due to its proximity to the Andes; its climate is classified as subtropical highland with dry winters and cool summers (Cwb). Measurements were recorded from 2013 to 2020 using an Eppley PSP pyranometer.

The Lq station, located in La Quiaca, is characterized by a cold semi-arid Andean steppe climate (BSk) and is notable for having some of the highest annual sunshine hours in Argentina. It was equipped with a Kipp & Zonen CMP11 pyranometer, with data collected from 2021 to 2023. The Yu station, situated in a humid subtropical region (Cwa), collected data from 2017 to 2018 using a CMP11 sensor. The Sca station, with a Cwb climate, recorded data from 2012 to 2013 using a CMP3 pyranometer. Finally, the Ero station, located in a high-altitude BSk region, collected data from 2016 to 2018, also using a CMP3 pyranometer.

All stations are equipped with pyranometers compliant with ISO 9060:2018 Class A or B standards for measuring global horizontal irradiance (GHI). Data were recorded at 1-minute intervals, with each value representing the average of six instantaneous samples taken every 10 seconds.

Table I: Stations considered in this work, with location, altitude, and climate classification.

| ID  | City       | Latitude | Longitude | Altitude (m) | Climate |
|-----|------------|----------|-----------|--------------|---------|
| Yu  | Yuto       | -23.58   | -64.50    | 401          | Cwa     |
| Sa  | Salta      | -24.72   | -65.40    | 1233         | Cwb     |
| Sca | San Carlos | -25.89   | -65.92    | 1624         | Cwb     |
| Er  | El Rosal   | -24.39   | -65.76    | 3355         | BSk     |
| Lq  | La Quiaca  | -22.10   | -65.60    | 3468         | BSk     |

The one-minute GHI measurements were subjected to the quality control procedure defined in Ref. [27]. The full procedure requires diffuse horizontal irradiance, but since this study only works with the GHI, a shorter version was used. Table II describes the filters used, where  $E$  is the solar constant,  $S$  is the Earth-Sun distance correction factor,  $\theta_z$  is the solar zenith

angle, and  $kt$  is the dimensionless clearness index, defined as the ratio of the GHI to the horizontal irradiance at the top of the atmosphere.

The measurement series were analyzed using the Standard Normal Homogeneity Test (SNHT) [28] to determine if the series displayed any changes in its behaviour resulting from an alteration or deviation in its normal trend. No alterations were detected in the series with p-values between 0.8 and 0.9.

Table II: Quality control filters applied to the measurements.

| Filter | Description                                                                          |
|--------|--------------------------------------------------------------------------------------|
| F1     | $GHI < 1.5 E S (\cos(\theta_z))^{1.2} + 100 \text{ W/m}^2$                           |
| F2     | $GHI > (6.5331 - 0.065502 \theta_z + 1.8312E-4 \theta_z^2) / (1 + 0.01113 \theta_z)$ |
| F3     | $kt < 1.4 \text{ \& } (90 - \theta_z) < 10^\circ$                                    |

The quality control procedure to Tab. II was applied to the ground-measured data, and one year of hourly averages was selected as the calibration dataset for model adjustment, while the remaining available data were used for validation purposes.

### III. MODELS

Two satellite-based and two reanalysis-based estimation models for GHI are assessed. These models are described in this section. The GHI estimates can be freely downloaded from respective online portals. A local implementation of cloud index satellite model [29], [30] is also included.

#### A. CAMS Heliosat-4

Heliosat-4 [31] is a fully physical model that employs a rapid yet precise approach to the libRadtran radiative transfer model [32]. This model utilizes various satellite data and the reanalysis database from the Copernicus Atmosphere Monitoring Service (CAMS). Cloud properties are derived from Meteosat Second Generation (MSG) satellite images at a 15-minute time rate using an adapted APOLLO scheme (AVHRR3 Processing Scheme Over Clouds, Land, and Ocean) [33]. The operational version of Heliosat-4 takes the form of lookup tables, enabling rapid computation. This model has already been evaluated in the region in studies such as [18], [19]; however, these studies do not report comparisons with other modeled solar estimation data sets. This dataset was downloaded from the SoDa website with a time resolution of 15 minutes. The MSG pixel size over the region is about 16 km.

#### B. LSA-SAF MDSSFTD

The MSG Downwelling Surface Short-wave Radiation Fluxes - Total and Diffuse (MDSSFTD) product is an instantaneous (every 15 minutes) estimate of the global and diffuse downward shortwave radiation flux at the surface. The method for retrieving this product consists of two separate modules: one for clear sky and one for cloudy sky conditions [34], [35]. The clear sky product uses the approach described in Ref. [36]. This method, known as SIRAMix (Surface Incident Radiation using Aerosol Mixtures), is a physical parameterization method coupled to a pre-calculated lookup table (LUT) of

aerosol properties, consisting mainly of the direct and diffuse transmittances of the different aerosol components and their corresponding aerosol albedos. The LUT is generated using radiative transfer models to vary aerosol loading, water vapor, and solar zenith angle. The cloudy GHI retrieval method uses the total shortwave cloud albedo at the top of the atmosphere (TOA) obtained from the observed TOA reflectances of the MSG. It uses the same aerosol estimates as in the clear sky case, and cloud-related terms are determined using a simplified radiative transfer model, similar to the method in Ref. [37]. Finally, these transmittances, namely the effective clear-sky transmittance estimated using gases and aerosols and the cloud transmittance estimated using the simplified radiative transfer model, are used together to determine the GHI.

This model has an approximate spatial resolution of 3 km with a 15-minute update frequency, and its estimates can be downloaded in NetCDF format from <https://datasaf.isasvcs.ipma.pt/PRODUCTS/MSG/MDSSFTD/>.

#### C. MERRA-2

The *Modern-Era Retrospective Analysis for Research and Applications, version 2* (MERRA-2) [38] is a reanalysis dataset developed by NASA with the Goddard Earth Observing System (GEOS-5.12.4) atmospheric data assimilation system [39], [40]. It is used to estimate various atmospheric parameters, including the downward shortwave irradiance, the GHI. The dataset covers the period from 1980 to the present with a latitude-longitude spatial resolution of  $0.5 \times 0.625^\circ$  and a time step of 60 minutes. This model improves the modeling of aerosols and related information affecting clear sky solar irradiance from the previous MERRA version 1 [41]. It also includes other physical improvements, such as water vapor content and cloud modeling [42], that affect solar irradiance estimates under all sky conditions. As a result of these updates, the dataset is expected to provide better solar irradiance performance than its predecessor. The data are freely available at <https://disc.gsfc.nasa.gov/datasets?project=MERRA-2>.

#### D. ERA-5

ERA5 is the fifth-generation atmospheric reanalysis model developed by the European Centre for Medium-Range Weather Forecasts (ECMWF), covering the period from January 1940 to the present. This model provides hourly estimates of numerous climate variables, including atmospheric, terrestrial, and oceanic aspects. The data are presented on a horizontal grid of 30 km, and the atmosphere is binned using 137 levels ranging from the Earth's surface to an altitude of 80 km [43]. The model is based on the Integrated Forecasting System (IFS) Cy41r2, which was operational until 2016. Thus, ERA5 benefits from a decade of advances in model physics, core dynamics, and data assimilation compared to the previous ERA-Interim data set. In addition to a significantly improved horizontal resolution of  $0.25^\circ$  latitude (31 km), ERA5 provides hourly output. This model combines satellite and in-situ observations in its estimates using advanced modeling and data assimilation techniques; therefore, ERA5 has been shown to produce more accurate

simulations compared to previous generations [44], [45]. The data are freely available at <https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels?tab=download>.

#### IV. METHODS

Simple linear regression models are evaluated for the various estimates available on an hourly scale. This requires temporal integration of ground measurements and model estimates originally obtained with an intrahour frequency (Heliosat-4, LSA-SAF). This section describes the performance metrics used and the various approaches taken to perform the regressions.

##### A. Performance Metrics

The most common performance indicators in the field of solar resource assessment have been covered by Ref. [46]; these include the Mean Bias Error (MBE), Mean Absolute Error (MAE), Root Mean Square Error (RMSE). The three metrics are defined as follows,

$$\text{MBE} = \frac{\sum_{i=1}^n (y_i - x_i)}{n}, \quad (1)$$

$$\text{MAE} = \frac{\sum_{i=1}^n |y_i - x_i|}{n}, \quad (2)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - x_i)^2}, \quad (3)$$

where  $x$  and  $y$  are the measured and estimated values, respectively, and  $n$  is the sample size. The MBE measures the systematic bias that a model can introduce in a long-term evaluation, while the MAE and RMSE measure the dispersion of the error using absolute and quadratic norms, respectively. Because of its greater sensitivity to outliers, the RMSE is often used in this area. Both dispersion metrics are reported here for completeness. The three indicators are presented in relative terms as a percentage of the mean of the measured values, referred to here as MBE (%), MAE (%), RMSE (%).

##### B. Approache 1: The traditional site adaptaion

The traditional approach is defined as follows,

$$GHI_{\text{adapted}} = a \cdot GHI_{\text{modeled}} + b, \quad (4)$$

##### C. Approach 2: Adaptation Based on a Error Series

In this approach, the regressor variable is the modeled GHI ( $GHI_{\text{modeled}}$ ), and the target variable is defined as the difference between the measured and modeled GHI, that is,  $\Delta_{GHI} = GHI_{\text{measured}} - GHI_{\text{modeled}}$ . Once the target variable is obtained, the adapted GHI is recovered by adding the error factor  $\Delta_{GHI}$  to the modeled GHI.

$$\Delta_{GHI} = GHI_{\text{measured}} - GHI_{\text{modeled}} \quad (5)$$

$$GHI_{\text{adapted}} = GHI_{\text{modeled}} + (a \cdot \Delta_{GHI} + b) \quad (6)$$

##### D. Approach 3: Adaptation Based on Error deseasonalized series

In this approach, the regressor variable is defined as the difference between the deseasonalized modeled GHI, which is obtained by subtracting the clear-sky GHI from the modeled GHI ( $GHI_{\text{modeled}} - GHI_{\text{clear sky}}$ ). The target variable corresponds to the difference between the deseasonalized measured GHI and the clear-sky GHI ( $GHI_{\text{measured}} - GHI_{\text{clear sky}}$ ). Once the target variable is estimated, the corrected GHI value is recovered by adding back the clear-sky GHI.

$$X = GHI_{\text{modeled}} - GHI_{\text{clear sky}} \quad (7)$$

$$Y = GHI_{\text{measured}} - GHI_{\text{clear sky}} \quad (8)$$

$$GHI_{\text{adapted}} = GHI_{\text{clear sky}} + (a \cdot Y + b) \quad (9)$$

In each case, the coefficients  $a$  and  $b$  are determined through simple linear regression.

#### V. RESULTS

Performance results are presented in Table III. Performance metrics are reported for each approach applied to all models.

Table III presents the performance metrics—Mean Bias Error (MBE), Mean Absolute Error (MAE), and Root Mean Square Error (RMSE)—for the CAMS, LSA-SAF, ERA-5, and MERRA-2 models under four site-adaptation conditions across the five locations described in Table I. The values are expressed in percentage terms, normalized by the mean of the measured GHI at each site. Table IV shows the coefficients  $a$  and  $b$  obtained for each approach.

Without site adaptation, all models exhibit substantial errors, particularly at the lower-altitude sites Yuto (Yu) and Salta (Sa). For instance, MERRA-2 reaches RMSE values of 52.4% at Ya and 65.3% at Sa, with strong positive bias. ERA-5 also shows considerable RMSE at these sites, accompanied by high MAE values. In contrast, CAMS and LSA-SAF display relatively lower RMSEs, typically in the 20–30% range, although with moderate to significant bias in several cases.

When applying Approach 1 (the traditional linear correction), performance improves modestly across most models and sites. MERRA-2 sees a notable reduction in RMSE at Sa, from 65.3% to 45.1%, while CAMS and LSA-SAF show more moderate changes. However, in some cases, MAE increases slightly, suggesting that this approach may introduce local distortions despite reducing bias.

Approach 2, which is based on the modeled error series, produces a more uniform and marked improvement in model accuracy. At Yu and Sa, RMSE values decrease substantially across all models. For example, ERA-5 at SA drops from 49.5% to 28.5% in RMSE, and similar behavior is observed for MERRA-2 and LSA-SAF. Moreover, MBE values tend to approach zero, and MAE is generally reduced, indicating improved alignment between modeled and measured GHI.

Approach 3, which uses deseasonalized error series, yields the lowest RMSE values in nearly all scenarios. At high-altitude sites such as El Rosal (Ero) and La Quiaca (Lq), CAMS and LSA-SAF reach RMSEs below 21%. MAE also decreases notably across most cases, and MBE values remain

Table III: Multi-approach performance assessment. The mean value of the normalization for the MBE, the MAE and the RMSE is of 429.2 W/m<sup>2</sup> for Yu, 432.9 W/m<sup>2</sup> for Sa, 554.4 W/m<sup>2</sup> for Sca, 628.4 W/m<sup>2</sup> for Ero and of 648.7 W/m<sup>2</sup> for Lq.

| Model   | Without Site Adaptation |        |         | Approache 1 |        |         | Approache 2 |        |         | Approache 3 |        |         |
|---------|-------------------------|--------|---------|-------------|--------|---------|-------------|--------|---------|-------------|--------|---------|
|         | MBE(%)                  | MAE(%) | RMSE(%) | MBE(%)      | MAE(%) | RMSE(%) | MBE(%)      | MAE(%) | RMSE(%) | MBE(%)      | MAE(%) | RMSE(%) |
| Yu      |                         |        |         |             |        |         |             |        |         |             |        |         |
| CAMS    | ≈ 0                     | 15.9   | 25.1    | -0.6        | 15.4   | 24.2    | -0.6        | 15.4   | 24.2    | -1.5        | 15.9   | 25.1    |
| LSA-SAF | 7.3                     | 15.4   | 24.7    | -5.4        | 17.3   | 24.1    | -0.5        | 15.5   | 24.7    | -5.5        | 15.8   | 24      |
| ERA-5   | 0.7                     | 22.9   | 34.5    | 2.4         | 23.1   | 34.5    | -1.1        | 15.6   | 25      | 0.2         | 24     | 33.2    |
| MERRA-2 | 24.7                    | 34.4   | 52.4    | -3          | 35.5   | 45.1    | -0.4        | 15.1   | 24      | -5          | 36.5   | 45.3.   |
| Sa      |                         |        |         |             |        |         |             |        |         |             |        |         |
| CAMS    | 3.6                     | 21.1   | 29.6    | -0.9        | 18.8   | 27.3    | -0.9        | 18.8   | 27.3    | -1.6        | 20.5   | 29.2    |
| LSA-SAF | 16.1                    | 24.3   | 34.2    | -1.4        | 21.6   | 29.8    | -0.9        | 19     | 27.6    | -2.4        | 20.9   | 30.2    |
| ERA-5   | 7.9                     | 40.2   | 49.5    | 0.8         | 38.6   | 46.6    | -1.1        | 19.2   | 28.5    | 1.0         | 34.7   | 44      |
| MERRA-2 | 44.6                    | 48.6   | 65.3    | 3.7         | 35.9   | 45.1    | -1.7        | 19     | 28.1    | 3.2         | 36.9   | 47.6    |
| Sca     |                         |        |         |             |        |         |             |        |         |             |        |         |
| CAMS    | 3.6                     | 24.2   | 31.8    | 0.1         | 19.5   | 26.9    | 0.1         | 19.5   | 26.9    | 7.6         | 20.7   | 29.8    |
| LSA-SAF | 16.4                    | 26.1   | 34.4    | 5.2         | 19.2   | 28.5    | -1.4        | 20.2   | 27.7    | 9.8         | 21.3   | 30.8    |
| ERA-5   | 4.4                     | 25.2   | 34.6    | -4.3        | 23.5   | 32.9    | 1.3         | 21.3   | 29.6    | 4.6         | 21.7   | 31.2    |
| MERRA-2 | 10.5                    | 25.4   | 34.1    | -2.8        | 23.3   | 31.1    | 1           | 21.3   | 29.6    | 5.1         | 22.9   | 32      |
| Ero     |                         |        |         |             |        |         |             |        |         |             |        |         |
| CAMS    | -20.2                   | 25.0   | 37.2    | 8.4         | 25.3   | 30      | 8.4         | 25.3   | 30      | 6.2         | 11.8   | 21.3    |
| LSA-SAF | -5.1                    | 14.3   | 23.8    | 5.5         | 17.3   | 23.7    | 9.2         | 27.7   | 32.4    | 5.4         | 11.1   | 20.4    |
| ERA-5   | -12.2                   | 20.3   | 26.6    | 5.1         | 18.5   | 24.2    | 9.2         | 27.8   | 35.5    | 5.3         | 12.1   | 21.3    |
| MERRA-2 | -1.5                    | 13.6   | 21.9    | 3.2         | 14.1   | 21.9    | 9           | 27.6   | 32.7    | 4.2         | 11.4   | 20.5    |
| Lq      |                         |        |         |             |        |         |             |        |         |             |        |         |
| CAMS    | -6                      | 14.2   | 21.6    | 2.1         | 13.5   | 19.4    | 2.1         | 13.5   | 19.4    | 0.5         | 10.2   | 16.2    |
| LSA-SAF | 3.4                     | 10.3   | 17.8    | -0.6        | 10.6   | 17.4    | 2.6         | 14.6   | 20.3    | -0.9        | 10.7   | 16.9    |
| ERA-5   | -3.1                    | 14.3   | 19.5    | 1           | 14.1   | 19.3    | 2.4         | 14.8   | 20.7    | -0.1        | 11.3   | 17      |
| MERRA-2 | 0.2                     | 12.7   | 20.1    | 0.5         | 12.8   | 19.9    | 2.4         | 14.6   | 20.6    | -0.6        | 11.4   | 17.9    |

close to zero. In these locations, all models achieve their best performance under Approach3.

Across all strategies, a clear trend is observed in relation to site altitude. Locations with higher elevation (Ero and Lq) generally present lower RMSE values, even without adaptation. Nevertheless, the improvement brought by the adaptation strategies—especially Approaches 2 and 3—is consistent across all sites, with particularly strong impact on the most error-prone models and conditions.

Figure 1 presents the mean bias error (MBE%) values for five sites located at different altitudes, using four models for estimating global horizontal irradiance (GHI), both before and after applying three site-adaptation approaches. Without adaptation the models exhibit significant systematic errors in certain cases, with MERRA-2 standing out due to MBE values exceeding 25% in Yu and 45% in Sa, indicating considerable overestimation. Following the application of the first adaptation approach—a direct linear correction applied to the modeled GHI—a substantial reduction in bias is observed across all models and sites. In Approaches 2 and 3, which introduce adjustments based on residual errors and deseasonalization, respectively, the MBE remains close to zero in most cases, indicating a systematic and robust improvement. This behavior is particularly evident in the high-altitude sites (Ero and Lq), where initial errors without adaptation were negative (underestimations), and all three methodologies successfully

correct this bias. Overall, the evidence suggests that site adaptation, regardless of the approach used, significantly enhances the accuracy of the models across the different geographic contexts evaluated.

Figure 2 illustrates the root mean square error (RMSE, %) performance of the CAMS, LSA-SAF, ERA-5, and MERRA-2 models under different site-adaptation strategies across five locations with varying altitudes. In the panel corresponding to the non-adapted condition (No Adapt.), high RMSE values are observed at lower-altitude sites, particularly at Sa and Yu, where MERRA-2 and ERA-5 exceed error rates of 50% and 45%, respectively; these results are consistent with findings reported in [24] and in other regions of South America such as [11]. This behavior contrasts with the higher-altitude sites (Ero and Lq), where RMSE tends to be lower even in the absence of adaptation. The implementation of Approach 1 leads to a significant reduction in RMSE for all models, with the most notable improvements observed in MERRA-2, especially at Sa and Yu, where errors decrease by more than 10 percentage points. Under Approach 2, all models exhibit converging RMSE values, generally remaining below 35% across all locations, indicating a systematic performance improvement. Approach 3 yields the lowest RMSE values at nearly all sites, with CAMS and LSA-SAF achieving errors below 20% at Lq and Ero. These results suggest that, while all adaptation strategies enhance model accuracy, Approach 3

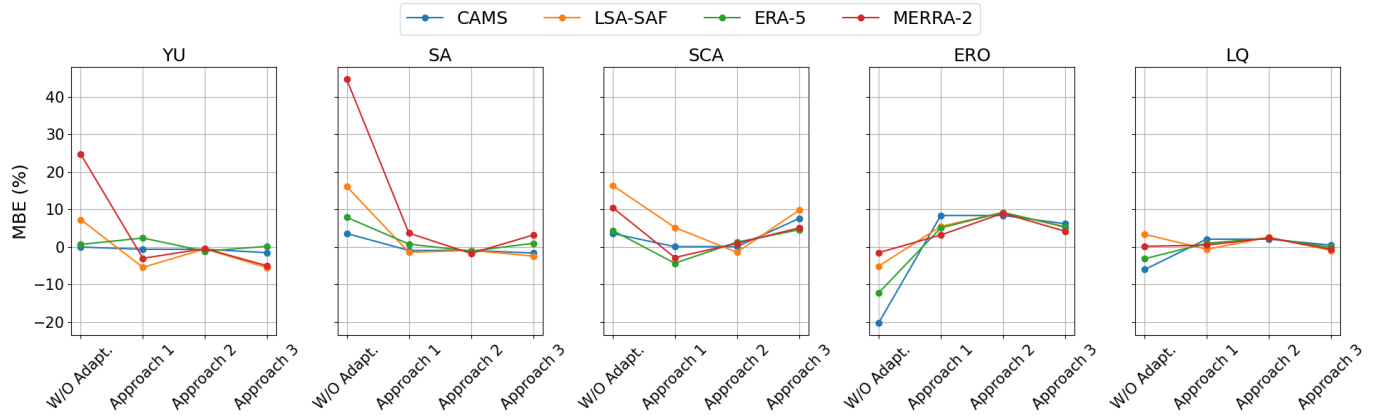


Figure 1: Mean Bias Error (MBE%) for stations (Yu, Sa, Sca, Ero, and Lq), comparing the performance of four irradiance models (CAMS, LSA-SAF, ERA-5, and MERRA-2). The x-axis displays the results without site-specific adaptation (W/O Adapt.) followed by three different adaptation approaches (Approach 1, 2, and 3), all based on linear adjustment methods. This figure illustrates the reduction of systematic bias in global horizontal irradiance (GHI) estimation achieved by applying different site-adaptation strategies.

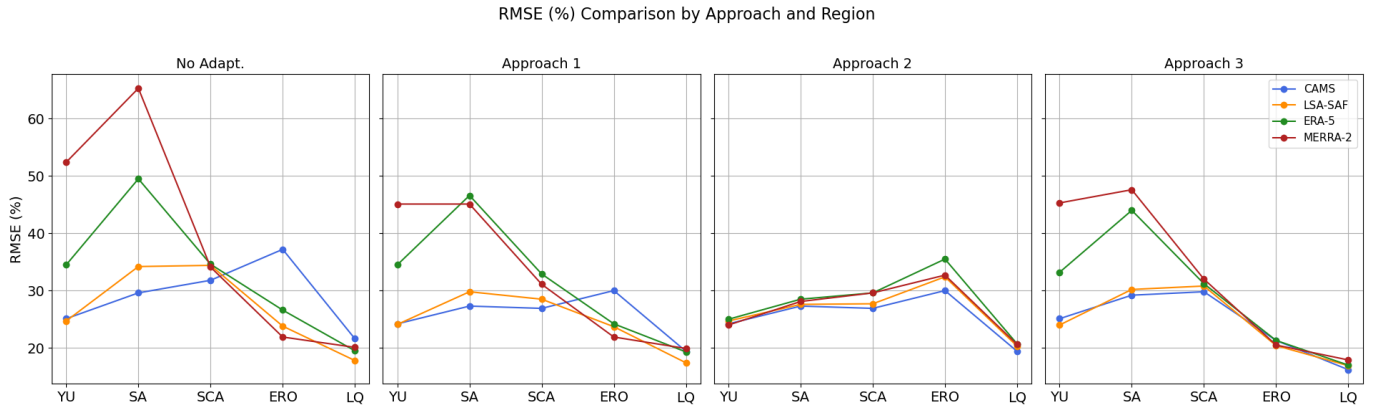


Figure 2: Comparison of the percentage root mean square error (RMSE, %) of the CAMS, LSA-SAF, ERA-5, and MERRA-2 models under different site-adaptation approaches across the five locations.

consistently provides the greatest RMSE reduction, regardless of model type or site altitude.

## VI. DISCUSSION

### A. Reanalysis vs Satelital models

The performance evaluation reveals consistent differences between reanalysis-based models (ERA-5 and MERRA-2) and satellite-based products (CAMS and LSA-SAF), particularly in their response to site-specific adaptation strategies and in raw accuracy before adaptation.

Without any adaptation, satellite-based models generally outperform their reanalysis-based counterparts across most sites. CAMS and LSA-SAF exhibit lower RMSE and MAE values at all locations, especially in high-altitude regions such as El Rosal (Ero) and La Quiaca (Lq), where atmospheric conditions are more stable and satellite-based retrievals benefit from clearer sky conditions. CAMS, in particular, maintains relatively low RMSE values in both lowland and highland sites, demonstrating robustness across different climate zones.

Conversely, reanalysis-based models—especially MERRA-2—tend to exhibit significantly higher errors in their native form, most notably at low-altitude, subtropical sites such as Yuto (Yu) and Salta (Sa). These elevated RMSE and MBE values suggest a misrepresentation of local radiative processes, likely due to the coarse spatial resolution and the physical parameterizations used in global reanalysis schemes. ERA-5 performs better than MERRA-2 in some locations but still lags behind satellite-based models in raw performance.

The application of site-adaptation techniques narrows the performance gap. Under Approaches 2 and 3, which incorporate error modeling and deseasonalization, ERA-5 and MERRA-2 show substantial reductions in RMSE and MAE, in some cases achieving comparable accuracy to CAMS and LSA-SAF. This result underscores the potential of post-processing strategies to correct for systematic biases inherent in reanalysis data, which, despite being physically consistent, often fail to capture local variability due to their coarser temporal and spatial resolutions.

Notably, MERRA-2 benefits the most from adaptation, par-

Table IV: Coeficientes obtenidos para cada enfoque.

|         | Approache 1 |         | Approache 2 |         | Approache 3 |          |
|---------|-------------|---------|-------------|---------|-------------|----------|
|         | a           | b       | a           | b       | a           | b        |
| Yu      |             |         |             |         |             |          |
| CAMS    | 0.74        | 129.606 | -0.259      | 129.606 | 0.456       | -94.986  |
| LSASAF  | 0.826       | 49.818  | -0.261      | 140.753 | 0.545       | -103.762 |
| ERA-5   | 0.83        | 49.572  | -0.203      | 105.177 | 0.482       | -109.643 |
| MERRA-2 | 0.83        | 29.417  | -0.204      | 101.619 | 0.505       | -119.032 |
| Sa      |             |         |             |         |             |          |
| CAMS    | 0.855       | 44.961  | -0.144      | 44.961  | 0.944       | -33.7    |
| LSASAF  | 0.921       | -37.025 | -0.153      | 57.394  | 1.093       | -67.854  |
| ERA-5   | 0.718       | 102.381 | -0.141      | 45.287  | 0.501       | -116.052 |
| MERRA-2 | 0.737       | -12.663 | -0.132      | 59.44   | 0.643       | -184.59  |
| Sca     |             |         |             |         |             |          |
| CAMS    | 0.74        | 129.606 | -0.259      | 129.606 | 0.464       | -94.986  |
| LSASAF  | 0.826       | 49.818  | -0.261      | 140.753 | 0.545       | -103.762 |
| ERA-5   | 0.83        | 49.572  | -0.203      | 105.177 | 0.482       | -109.643 |
| MERRA-2 | 0.83        | 29.417  | -0.204      | 110.619 | 0.505       | -119.032 |
| Ero     |             |         |             |         |             |          |
| CAMS    | 0.777       | 291.517 | -0.222      | 291.517 | 0.216       | 3.1      |
| LSASAF  | 0.963       | 89.297  | -0.031      | 203.866 | 0.394       | -2.163   |
| ERA-5   | 1.033       | 91.417  | 0.008       | 180.476 | 0.342       | 6.35     |
| MERRA-2 | 0.967       | 49.978  | 0.106       | 117.95  | 0.535       | -6.16    |
| Lq      |             |         |             |         |             |          |
| CAMS    | 0.847       | 145.941 | -0.152      | 145.941 | 0.51        | -17.416  |
| LSASAF  | 0.952       | 4.836   | -0.099      | 122.508 | 0.7199      | -45.696  |
| ERA-5   | 0.979       | 40.172  | -0.072      | 100.892 | 0.541       | -28.119  |
| MERRA-2 | 0.921       | 53.607  | -0.08       | 107.45  | 0.413       | -53.793  |

ticularly under Approach 2, where RMSE reductions of over 20 percentage points are observed in the worst-performing sites. This suggests that, while the initial performance is limited, reanalysis-based models can be significantly improved through targeted correction methods. In contrast, CAMS and LSA-SAF show less pronounced improvements with adaptation, as their baseline performance is already relatively high, especially in clear-sky and mountainous conditions.

Overall, the results highlight the intrinsic trade-off between physical consistency and spatial resolution. Satellite-based models offer more accurate short-term estimates of surface radiation under a wide range of topographic and climatic conditions, while reanalysis products, although initially less precise, can achieve competitive performance when coupled with suitable correction schemes.

### B. CAMS vs LSA-SAF

Although both CAMS and LSA-SAF use the same geostationary satellite platform (Meteosat-2) to obtain radiative data, performance evaluation reveals significant discrepancies in their ability to reproduce ground-level global horizontal irradiance (GHI) at different sites. This divergence is especially notable in high-altitude regions such as El Rosal (Ero), where CAMS exhibits a significantly high RMSE in its unadapted

form (37.2%), while LSA-SAF achieves notably lower error metrics under the same conditions (23.8%).

This contrast suggests that, beyond the shared satellite origin, differences in the processing chains, radiative transfer models, and data assimilation methods used by each system substantially influence model accuracy. CAMS integrates satellite data into a global atmospheric modeling framework that incorporates aerosol and atmospheric composition forecasts, which, while comprehensive, may not effectively capture local-scale radiative dynamics in complex terrain. This limitation is particularly evident in Ero, where orographic effects and local atmospheric patterns—characteristic of high-altitude, semi-arid regions—introduce variability that is difficult to resolve with approximate global models.

In contrast, LSA-SAF is specifically designed for the derivation of land surface variables from satellite imagery and employs retrieval algorithms more directly linked to local surface conditions. Its relative robustness in Ero may be due to these region-centric radiative transfer methodologies, which are more sensitive to topographic and albedo variations, resulting in better performance at sites with complex terrain and less atmospheric homogeneity.

Furthermore, adaptation strategies affect both products differently. While CAMS demonstrates substantial improvement when the proposed adaptation methods (Approaches 2 and 3) are applied, the relative benefit is more modest compared to LSA-SAF. This pattern implies that CAMS's reference errors may originate from structural aspects of its modeling framework, which are only partially correctable using linear or error-based adaptation techniques. On the other hand, LSA-SAF benefits moderately from adaptation, reinforcing its potential for applications where both satellite data availability and topographic sensitivity are crucial.

These findings emphasize that simply coming from a shared satellite sensor does not guarantee similar performance. Instead, model design, algorithmic structure, and intended operational purpose (global atmospheric prediction vs. land surface monitoring) significantly influence the final accuracy of GHI estimates. Therefore, users should be cautious when assuming equivalence between products such as CAMS and LSA-SAF, especially in regions with harsh climatic and geographical conditions.

### C. Site adaptation Approaches

The performance evaluation of the three site-adaptation strategies applied to the GHI estimations reveals distinct improvements across all models and sites, though with varying degrees of effectiveness. Each approach introduces a systematic correction to the original modeled irradiance, targeting different sources of model bias and local variability.

Approach 1, based on a traditional linear correction between modeled and measured GHI values, consistently reduces RMSE across all datasets when compared to the unadapted condition. This improvement is particularly notable in models with large initial errors, such as MERRA-2 and ERA-5 at low-altitude sites like Yuto (Yu) and Salta (Sa), where RMSE reductions exceeding 7–10 percentage points are observed.



However, while effective in bias mitigation, Approach 1 often fails to substantially reduce the MAE or fully capture the structure of local error dynamics, particularly in regions with greater climatic variability.

Approach 2, which adapts the modeled GHI based on error series derived from the difference between measured and modeled values, demonstrates a more balanced improvement across all error metrics (MBE, MAE, and RMSE). The normalization of RMSE values under this method tends to converge below 30% for most models and sites, including typically challenging scenarios such as MERRA-2 at Sa or ERA-5 at Sca. This suggests that modeling the local error behavior explicitly—rather than relying solely on a global correction factor—yields a more nuanced and effective adjustment, especially for reanalysis-based products.

Approach 3, which introduces deseasonalized error series by leveraging clear-sky GHI as a reference, delivers the most consistent and lowest RMSE values across almost all scenarios. Notably, in high-altitude sites such as El Rosal (Ero) and La Quiaca (Lq), models adapted using Approach 3 reach RMSE levels below 22% in several cases (e.g., CAMS, LSA-SAF). This suggests a clear advantage in decoupling the seasonal radiative cycle from stochastic error components, enabling more accurate modeling of residual variability. Additionally, this method appears especially beneficial for satellite-based models, where clear-sky estimations are directly linked to the radiative retrieval algorithms.

In summary, while all three approaches enhance model performance relative to the unadapted case, Approaches 2 and 3 show superior ability to generalize across varying topographies and model types. Approach 3, in particular, provides the most consistent error reduction and is best suited for complex climatic contexts due to its capacity to isolate and model non-seasonal discrepancies.

## VII. CONCLUSIONS

This study has evaluated the performance of four global horizontal irradiance (GHI) estimation models—two satellite-based (CAMS Heliosat-4 and LSA-SAF MDSSFTD) and two reanalysis-based (MERRA-2 and ERA-5)—across five representative sites in northwestern Argentina, applying three site-adaptation approaches. The main conclusions drawn from the results are as follows:

**Importance of Site Adaptation:** All models exhibited a significant improvement in accuracy following the application of site-adaptation methods. This highlights the necessity of adjusting modeled estimates using local measurements to minimize systematic biases and enhance data representativeness, particularly in geographically complex regions such as northwestern Argentina.

**Model Comparison:** Without adaptation, satellite-based models—particularly CAMS—outperformed reanalysis models, especially at higher-altitude locations. However, reanalysis models, especially MERRA-2, showed notable performance improvements when site-adaptation strategies were applied, achieving results comparable to those of the satellite-based models.

**Evaluation of Adaptation Approaches:** The traditional approach (Approach 1) generally reduced bias but did not substantially improve error dispersion. In contrast, Approaches 2 and 3, which are based on error series and deseasonalized errors, respectively, delivered more consistent improvements across all sites. Approach 3 achieved the lowest RMSE values, particularly in high-altitude locations.

**Performance by Altitude:** A clear trend was observed in which higher-altitude sites (El Rosal and La Quiaca) exhibited lower estimation errors, even without adaptation. This suggests greater atmospheric stability and more favorable conditions for satellite-based models at these elevations.

**Regional Relevance:** Given the topographic and climatic heterogeneity of northwestern Argentina, this work provides a valuable reference for the selection and calibration of irradiance models in the region. It also offers evidence of the effectiveness of simple site-adaptation methods, which can be implemented even with limited resources.

In summary, site adaptation proves to be a fundamental tool for improving solar resource estimation in regions with limited measurement infrastructure. Among the approaches considered, those incorporating error modeling and deseasonalization offer the most robust and generalizable improvements.

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